Lesson 8.4 - The Central Limit Theorem

> **Video:** Central Limit Theorem - Sampling Distribution of Sample Means by The Organic Chemistry Tutor
> **Chapter:** 8 - Normal Probability Distributions

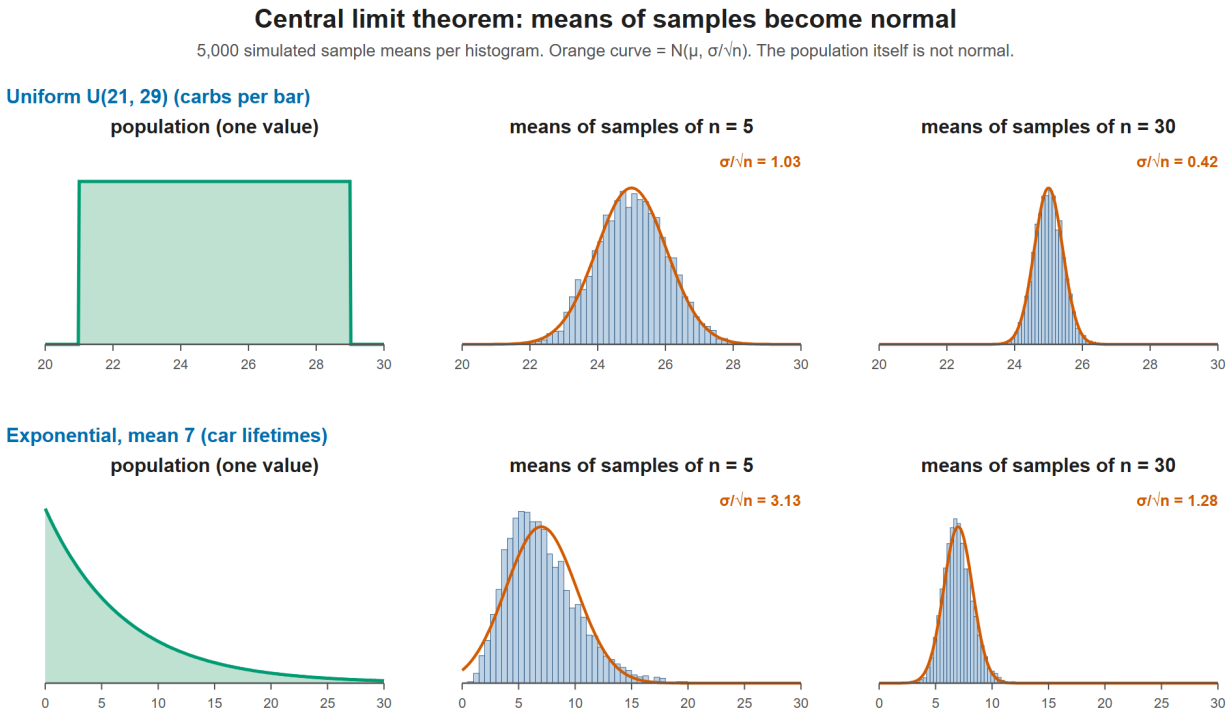
The idea

1. Take many samples of size n from a population.
2. Compute the mean $\bar{x}$ of each sample.
3. Plot all those sample means in a histogram.

That histogram is the **sampling distribution of the sample means**, and it is **approximately normal**:

Population shape	Sampling distribution of $\bar{x}$
Normal	Normal for **any** n
Anything else (uniform, exponential, skewed, irregular, ...)	Approximately normal when n is large enough. $n \geq 30$ is the usual rule of thumb

Example: in a town of 100,000, take repeated samples of 30 people and record each sample's mean age (41.8, 39.6, 40.5, ...). Those means form a bell curve even if individual ages do not.



Larger n: the histogram becomes more bell-shaped AND narrower, always centered on the population mean μ.

Neither population is normal, yet the sample means pile up into a bell shape. With n = 30 they match the normal curve N(μ, σ/√n) closely, even for the skewed exponential population, where n = 5 is still visibly lopsided.

Symbols

Symbol	Meaning
μ	Population mean
σ	Population standard deviation
$\bar{x}$	Mean of **one** sample
s	Standard deviation of **one** sample
$\mu_{\bar{x}}$	Mean of the **sampling distribution**: $\mu_{\bar{x}} = \mu$
$\sigma_{\bar{x}}$	SD of the sampling distribution, the **standard error**: $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$
n	Sample size

Law of large numbers link: as n grows, $\bar{x}$ gets closer to μ. For example, mean ages from samples of 10, 50, 100 and 1,000 people might be 32, 38.5, 35.3 and 36.1, approaching the true 36.

Larger n → narrower curve

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

n is in the denominator, so **a larger sample gives a smaller standard error**. The sampling distribution becomes **taller and narrower**: $n = 50$ is narrower than $n = 30$, which is narrower than $n = 10$.

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z-scores: individuals vs means vs sums

Question about...	Distribution	z-score
One value x	$X \sim N(\mu, \sigma)$	$z = \frac{x - \mu}{\sigma}$
The **mean** of n values	$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$	$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$
The **sum** of n values	$\sum X \sim N(n\mu, \sqrt{n}\,\sigma)$	$z = \frac{\sum x - n\mu}{\sqrt{n}\,\sigma}$

Then use the z-table (left area); right area = 1 − left area. To go from a percentile back to a value: $\bar{x} = \mu + z\,\frac{\sigma}{\sqrt{n}}$.

Reminders from earlier lessons

- **Uniform** $U(a, b)$: $f(x) = \frac{1}{b-a}$, $\mu = \frac{a+b}{2}$, $\sigma = \frac{b-a}{\sqrt{12}}$, probability = base × height.
- **Exponential**: $f(x) = \lambda e^{-\lambda x}$, $\lambda = 1/\mu$, $\sigma = \mu$, $P(X < x) = 1 - e^{-\lambda x}$.

Example 1 - entrance exam scores: normal, $\mu = 74$, $\sigma = 6.8$

(a) One student scores below 65 ($n = 1$, so use the population distribution)

$$z = \frac{x-\mu}{\sigma} = \frac{65-74}{6.8} \approx -1.32 \;\Rightarrow\; P = 0.09342 \approx 9.34\%$$

(b) The mean of 50 students is above 75

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{6.8}{\sqrt{50}} \approx 0.9617, \qquad z = \frac{\bar{x}-\mu}{\sigma_{\bar{x}}} = \frac{75-74}{0.9617} \approx 1.04$$

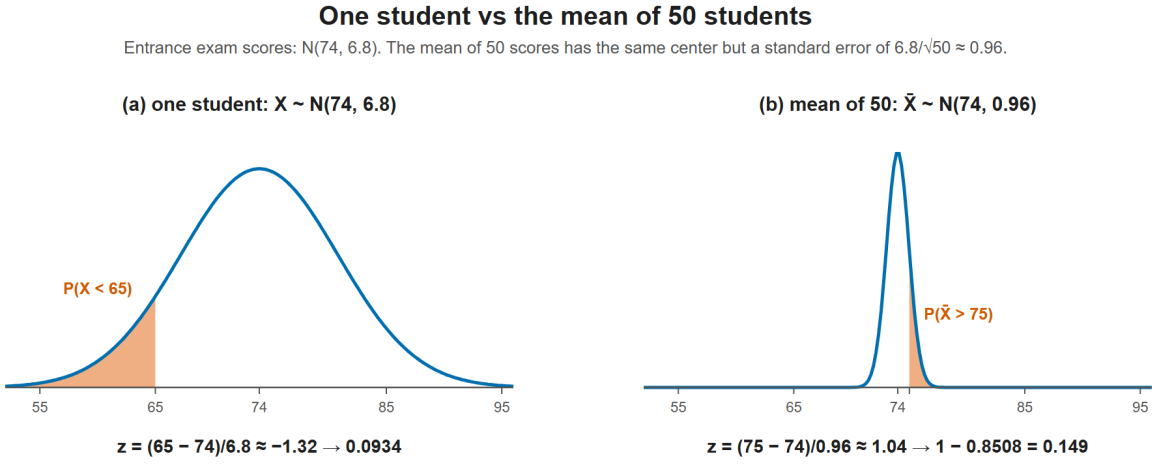
$$P(\bar{X} > 75) = 1 - P(z < 1.04) = 1 - 0.85083 = 0.14917 \approx 14.9\%$$

(c) Distribution of the mean of 50 scores

$$\bar{X} \sim N(74,\ 0.9617)$$

(d) 80th percentile of the mean score A left area of 0.80 gives $z \approx 0.84$ (table value 0.79955):

$$\bar{x} = \mu + z\,\sigma_{\bar{x}} = 74 + 0.84(0.9617) \approx 74.81$$



Example 1 on a shared axis. A question about one student uses σ = 6.8. A question about the mean of 50 students uses the much smaller standard error 6.8/√50 ≈ 0.96.

Example 2 - carbs in a snack bar: uniform from 21 g to 29 g, $n = 100$

(a) One bar

$$X \sim U(21, 29), \qquad \mu = \frac{a+b}{2} = \frac{21+29}{2} = 25, \qquad \sigma = \frac{b-a}{\sqrt{12}} = \frac{29-21}{\sqrt{12}} \approx 2.3094$$

(b) Mean of 100 bars. By the CLT ($n \geq 30$) it is **normal**, even though one bar is uniform:

$$\bar{X} \sim N\left(\mu,\ \frac{\sigma}{\sqrt{n}}\right) = N\left(25,\ \frac{2.3094}{\sqrt{100}}\right) = N(25,\ 0.23094)$$

(c) One bar has 24-26 g (uniform, so base × height)

$$P(24 < X < 26) = \text{width} \times \frac{1}{b-a} = (26-24) \times \frac{1}{8} = 0.25$$

(d) The mean of 100 bars is between 24.9 g and 25.1 g (normal)

$$z = \frac{\bar{x}-\mu}{\sigma_{\bar{x}}} = \frac{25 \pm 0.1 - 25}{0.23094} = \frac{\pm 0.1}{0.23094} \approx \pm 0.43 \;\Rightarrow\; P = 0.66640 - 0.33360 = 0.3328$$

(e) Distribution of the total carbs in 100 bars

$$\sum X \sim N\big(n\mu,\ \sqrt{n}\,\sigma\big) = N\big(100 \cdot 25,\ 10 \cdot 2.3094\big) = N(2500,\ 23.094)$$

(f) The total is more than 2,540 g

$$z = \frac{\sum x - n\mu}{\sqrt{n}\,\sigma} = \frac{2540 - 2500}{23.094} \approx 1.73 \ \Rightarrow\ P = 1 - 0.95818 = 0.04182 \approx 4.2\%$$

Example 3 - car lifetimes: exponential with mean 7 years, $n = 40$ cars

(a) Rate and SD (one car)

$$\lambda = \frac{1}{7} \text{ per year}, \qquad \sigma = \mu = 7 \text{ years}$$

(b) Distribution of the mean lifetime of 40 cars (CLT: normal)

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{7}{\sqrt{40}} \approx 1.1068, \qquad \bar{X} \sim N(7,\ 1.1068)$$

(c) P(sample mean < 6.5 years)

$$z = \frac{\bar{x} - \mu}{\sigma_{\bar{x}}} = \frac{6.5 - 7}{1.1068} \approx -0.45 \ \Rightarrow\ P = 0.32636 \approx 32.6\%$$

(d) IQR of the sample mean

- Q_1 (left area 0.25): the table gives $z = -0.67 \to 0.2514$ and $z = -0.68 \to 0.2483$, so use $z \approx -0.675$.

$$Q_1 = 7 - 0.675(1.1068) \approx 6.253$$

- Q_3 (left area 0.75): $z \approx +0.675$

$$Q_3 = 7 + 0.675(1.1068) \approx 7.747$$

- $$\text{IQR} = 7.747 - 6.253 \approx 1.494 \text{ years}$$

Key takeaways

- Sample means are approximately **normal** for $n \geq 30$, whatever the population's shape (exactly normal if the population is normal).
- $\mu_{\bar{x}} = \mu$ and $\sigma_{\bar{x}} = \sigma/\sqrt{n}$, so a bigger n gives a smaller spread.
- Decide whether the question is about **one** value (use σ and the population's own shape), a **mean** (use $\sigma/\sqrt{n}$, normal), or a **sum** (use $n\mu$ and $\sqrt{n}\,\sigma$, normal).